



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to SS AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

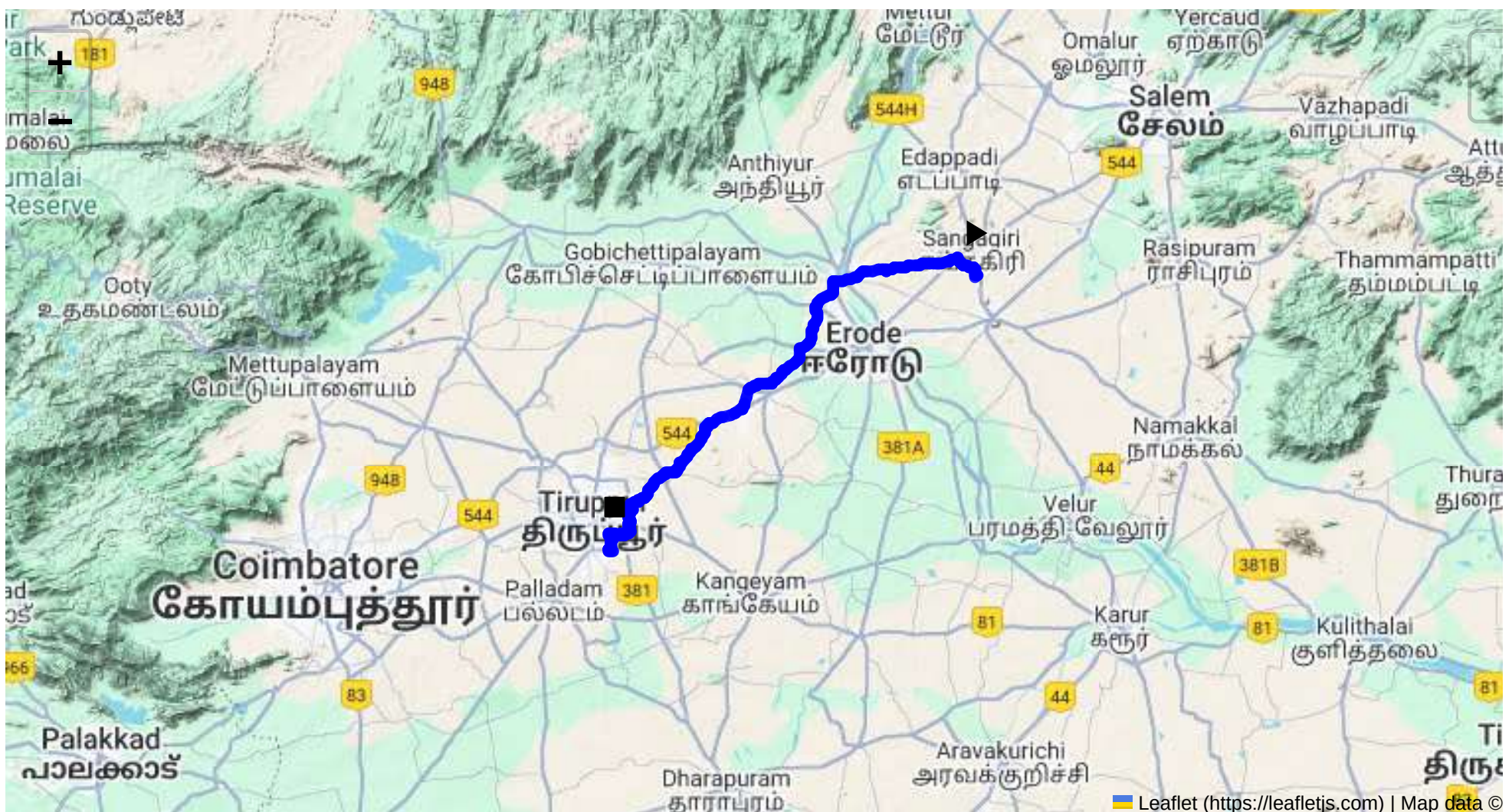
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 87.56 km**  
**Estimated Duration: 1.9 hours**  
**Adjusted Duration (Heavy Vehicle): 2.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.062385, 77.368607)**



## Welcome to the Journey Risk Management Study

### Route Safety Analysis Report

**1. Overview of the Route Map:** The route from Sangagiri to K Chettipalayam typically follows the NH544 and SH19, spanning approximately 87.56 kilometers. This is a key route in Tamil Nadu connecting textile hubs and industrial areas. It traverses through a combination of urban, semi-urban, and rural settings, mostly linear with moderate curves.



**2. Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu generally experiences tropical weather. The region can be prone to high temperatures, particularly from March to June, and receives significant rainfall during the monsoon seasons (June to September and October to December). Heavy rain can lead to waterlogging and reduced visibility. Occasionally, cyclonic storms may affect the area, causing severe disruption.

**3. Analysis of Traffic Patterns:**

- **Peak Hours:** Traffic tends to be heavier during morning (8:00 AM - 10:30 AM) and evening (5:00 PM - 8:00 PM) rush hours.
- **Congestion-Prone Areas:** Potential congestion zones include intersections near Salem and Avinashi, and around the entrance to Tiruppur.
- It is advisable to plan travel during off-peak times or divert through rural roads with less congestion.

**4. Assessment of Road Quality and Infrastructure:**

- **NH544:** Generally, the highway is well-maintained with good signage and multiple lanes. However, notable wear and tear can occur due to heavy usage.
- **SH19:** Road quality varies, with some segments experiencing potholes and uneven surfaces. Periodic maintenance causes temporary lane closures and bottlenecks.

**5. Suggestions for Alternative Routes:** In case of emergencies or severe congestion, consider the Salem-Kumarapalayam-Kangayam-Tiruppur road as an alternative. Though slightly longer, it offers less traffic and comparable travel time.

**6. Summary of Local Regulations Affecting Hazardous Material Transport:**

- Truck drivers carrying hazardous materials must adhere to strict state and national regulations.
- Transport of hazardous materials is restricted during peak urban congestion times to mitigate risks.
- Comprehensive documentation and vehicle markings are mandatory.

**7. Overview of Historical Incidents:** The region has seen periodic road accidents involving heavy vehicles, often due to driver fatigue, speeding, or vehicle failure. Most incidents occur near urban entry points or where road conditions suddenly change.

**8. Environmental Considerations and Sensitive Areas:**

- Areas around Sangagiri are primarily agricultural; caution is advised to minimize impact on local farms.
- Awareness of natural habitats is necessary as the route crosses through regions with wildlife presence.

**9. Analysis of Communication Coverage:** Mobile network coverage is generally robust across the route due to its prominence. However, rural stretches, especially near agricultural areas, might experience temporary signal loss.

**10. Estimated Emergency Response Times:**

- Urban areas: 15-30 minutes due to proximity to facilities.

- Rural areas: 30-60 minutes, depending on accessibility and the specific location.
- Service responsiveness generally high near Salem and Tiruppur.

**11. Overall Summary of Risk Assessment:** Overall, the route is moderately safe but requires attention to traffic patterns and weather conditions. Ensuring vehicle readiness and compliance with safety regulations is critical for minimizing risk. Adequate planning, awareness of alternative routes, and understanding the local context significantly enhance safety for heavy vehicles transporting hazardous materials.

Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.44024, 77.87528 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44898, 77.87410 | 15 KM/Hr    |
| 3  | Turn       | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4  | Turn       | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.44965, 77.77565 | 30 KM/Hr    |
| 8  | Turn       | Medium     | 11.44953, 77.77344 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.43165, 77.67734 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.23724, 77.50686 | 30 KM/Hr    |
| 11 | Turn       | High       | 11.23704, 77.50676 | 15 KM/Hr    |
| 12 | Blind Spot | Blind Spot | 11.23686, 77.50216 | 10 KM/Hr    |
| 13 | Turn       | Medium     | 11.23649, 77.50219 | 30 KM/Hr    |
| 14 | Turn       | High       | 11.23396, 77.50105 | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.23287, 77.49934 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.18407, 77.46330 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.18273, 77.46224 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.17521, 77.45978 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.17368, 77.45730 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.17111, 77.45547 | 30 KM/Hr    |
| 21 | Turn       | High       | 11.16959, 77.44396 | 15 KM/Hr    |
| 22 | Turn       | Medium     | 11.16381, 77.43700 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 23 | Turn       | Medium     | 11.16280, 77.43139 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.15217, 77.42418 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.15134, 77.42300 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.15023, 77.42247 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.14925, 77.42065 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.13848, 77.41611 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.13578, 77.41299 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.12989, 77.40302 | 30 KM/Hr    |
| 31 | Turn       | High       | 11.12775, 77.39347 | 15 KM/Hr    |
| 32 | Turn       | High       | 11.11846, 77.39460 | 15 KM/Hr    |
| 33 | Turn       | High       | 11.11833, 77.39390 | 15 KM/Hr    |
| 34 | Turn       | High       | 11.10276, 77.39199 | 15 KM/Hr    |
| 35 | Turn       | High       | 11.10256, 77.39273 | 15 KM/Hr    |
| 36 | Blind Spot | Blind Spot | 11.09882, 77.39312 | 10 KM/Hr    |
| 37 | Turn       | High       | 11.09898, 77.39028 | 15 KM/Hr    |
| 38 | Turn       | Medium     | 11.09456, 77.39233 | 30 KM/Hr    |
| 39 | Turn       | High       | 11.08759, 77.39166 | 15 KM/Hr    |
| 40 | Turn       | Medium     | 11.08531, 77.38413 | 30 KM/Hr    |
| 41 | Turn       | Medium     | 11.08597, 77.38095 | 30 KM/Hr    |
| 42 | Turn       | High       | 11.08536, 77.37919 | 15 KM/Hr    |
| 43 | Turn       | Medium     | 11.08481, 77.37902 | 30 KM/Hr    |
| 44 | Turn       | Medium     | 11.08448, 77.37842 | 30 KM/Hr    |
| 45 | Turn       | Medium     | 11.08510, 77.37541 | 30 KM/Hr    |
| 46 | Turn       | Medium     | 11.08463, 77.37378 | 30 KM/Hr    |
| 47 | Turn       | High       | 11.08497, 77.37234 | 15 KM/Hr    |
| 48 | Turn       | High       | 11.08544, 77.37228 | 15 KM/Hr    |
| 49 | Turn       | High       | 11.08604, 77.36377 | 15 KM/Hr    |
| 50 | Turn       | High       | 11.08583, 77.36360 | 15 KM/Hr    |
| 51 | Turn       | High       | 11.07888, 77.36793 | 15 KM/Hr    |
| 52 | Turn       | Medium     | 11.07037, 77.36701 | 30 KM/Hr    |
| 53 | Turn       | Medium     | 11.06962, 77.36676 | 30 KM/Hr    |
| 54 | Turn       | Medium     | 11.06907, 77.36623 | 30 KM/Hr    |
| 55 | Turn       | Medium     | 11.06888, 77.36568 | 30 KM/Hr    |
| 56 | Blind Spot | Blind Spot | 11.06421, 77.36175 | 10 KM/Hr    |

## Emergency Locations

|    | type     | name                                           | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------------------|---------------------------|-------------|------------|
| 3  | hospital | Dhanvanthri Hospital                           | 11.4496712,<br>77.7593772 | 30 km/h     | Medium     |
| 4  | hospital | J.K.K. Trust Hospital                          | 11.4445841,<br>77.7307962 | 30 km/h     | Medium     |
| 6  | hospital | Shri Sathyanarayana Hospital                   | 11.4291297,<br>77.6913408 | 30 km/h     | Medium     |
| 7  | hospital | Thanish Siddha Hospital                        | 11.430003, 77.674964      | 30 km/h     | Medium     |
| 8  | clinic   | Harshitha Clinic                               | 11.4313207,<br>77.674718  | 30 km/h     | Medium     |
| 9  | hospital | Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam | 11.432369, 77.674894      | 30 km/h     | Medium     |
| 10 | clinic   | G.K Clinic                                     | 11.4297244,<br>77.6749715 | 30 km/h     | Medium     |
| 11 | clinic   | Erode Cancer Centre                            | 11.3732, 77.649152        | 30 km/h     | Medium     |
| 13 | hospital | Irt Hospital                                   | 11.2803603,<br>77.5644118 | 30 km/h     | Medium     |
| 15 | hospital | Gen Siddha Hospital                            | 11.2375033,<br>77.5059797 | 30 km/h     | Medium     |
| 16 | hospital | Vijayamangalm Government Hospital              | 11.2382827,<br>77.501687  | 30 km/h     | Medium     |
| 18 | hospital | Aravind Eye Care Hospital                      | 11.1729469,<br>77.4574467 | 30 km/h     | Medium     |
| 19 | clinic   | Sri Maha Annai Clinic                          | 11.1707162,<br>77.4536498 | 30 km/h     | Medium     |
| 20 | hospital | Ram Doctor Hospital                            | 11.1709391,<br>77.4532808 | 30 km/h     | Medium     |
| 21 | clinic   | Kumaran Clinic                                 | 11.1697553,<br>77.451543  | 30 km/h     | Medium     |
| 22 | hospital | Sri Aravindhar Hospital                        | 11.1697756,<br>77.4448276 | 30 km/h     | Medium     |
| 23 | clinic   | Dr. Raja Clinic                                | 11.1278464,<br>77.3927768 | 30 km/h     | Medium     |
| 24 | clinic   | D K Clinic General & Ortho                     | 11.1117864,<br>77.3906835 | 30 km/h     | Medium     |



|    | type     | name                          | coordinates              | speed_limit | risk_level |
|----|----------|-------------------------------|--------------------------|-------------|------------|
| 25 | hospital | Tiruppur, Government Hospital | 11.083827,<br>77.3634797 | 30 km/h     | Medium     |
| 26 | hospital | Meenakshi Hospital - Tiruppur | 11.07082, 77.369519      | 30 km/h     | Medium     |

### Crowded Spots

|    | type        | name                                      | coordinates               | speed_limit | risk_level |
|----|-------------|-------------------------------------------|---------------------------|-------------|------------|
| 0  | school      | KRP Matric. Hr. Sec School                | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 1  | college     | Vivekanandha Engineering College          | 11.4589312,<br>77.7899284 | 30 km/h     | Medium     |
| 2  | marketplace | Monday market                             | 11.452863,<br>77.775989   | 30 km/h     | Medium     |
| 5  | school      | SSM Matriculation Higher Secondary School | 11.4321653,<br>77.6880046 | 30 km/h     | Medium     |
| 12 | college     | Government polytechnic college            | 11.2907053,<br>77.5698726 | 30 km/h     | Medium     |
| 14 | school      | Bharathi Matriculation School             | 11.2494613,<br>77.5307028 | 30 km/h     | Medium     |
| 17 | college     | Sasurie Engineering College               | 11.2171482,<br>77.4898552 | 30 km/h     | Medium     |

### Route Photos of Risky Spots



Risk Type: Turn  
Risk Level: High



**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44024, 77.87528**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44898, 77.87410**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.45352, 77.85698**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45898, 77.85135





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44965, 77.77565



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.44953, 77.77344





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.43165, 77.67734



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.23724, 77.50686





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.23704, 77.50676



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.23686, 77.50216



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.23649, 77.50219





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.23396, 77.50105



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.23287, 77.49934





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.18407, 77.46330



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.18273, 77.46224



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.17521, 77.45978





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.17368, 77.45730



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.17111, 77.45547





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.16959, 77.44396



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16381, 77.43700



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16280, 77.43139





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.15217, 77.42418



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.15134, 77.42300





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.15023, 77.42247



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.14925, 77.42065



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.13848, 77.41611





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.13578, 77.41299



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.12989, 77.40302





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.12775, 77.39347



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.11846, 77.39460



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.11833, 77.39390





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.10276, 77.39199



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.10256, 77.39273





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.09882, 77.39312



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.09898, 77.39028



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.09456, 77.39233





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.08759, 77.39166



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08531, 77.38413





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08597, 77.38095



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.08536, 77.37919



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08481, 77.37902





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08448, 77.37842



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.08510, 77.37541





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08463, 77.37378



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.08497, 77.37234



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.08544, 77.37228





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.08604, 77.36377



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.08583, 77.36360





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.07888, 77.36793



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.07037, 77.36701



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.06962, 77.36676





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.06907, 77.36623



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.06888, 77.36568





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.06421, 77.36175

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